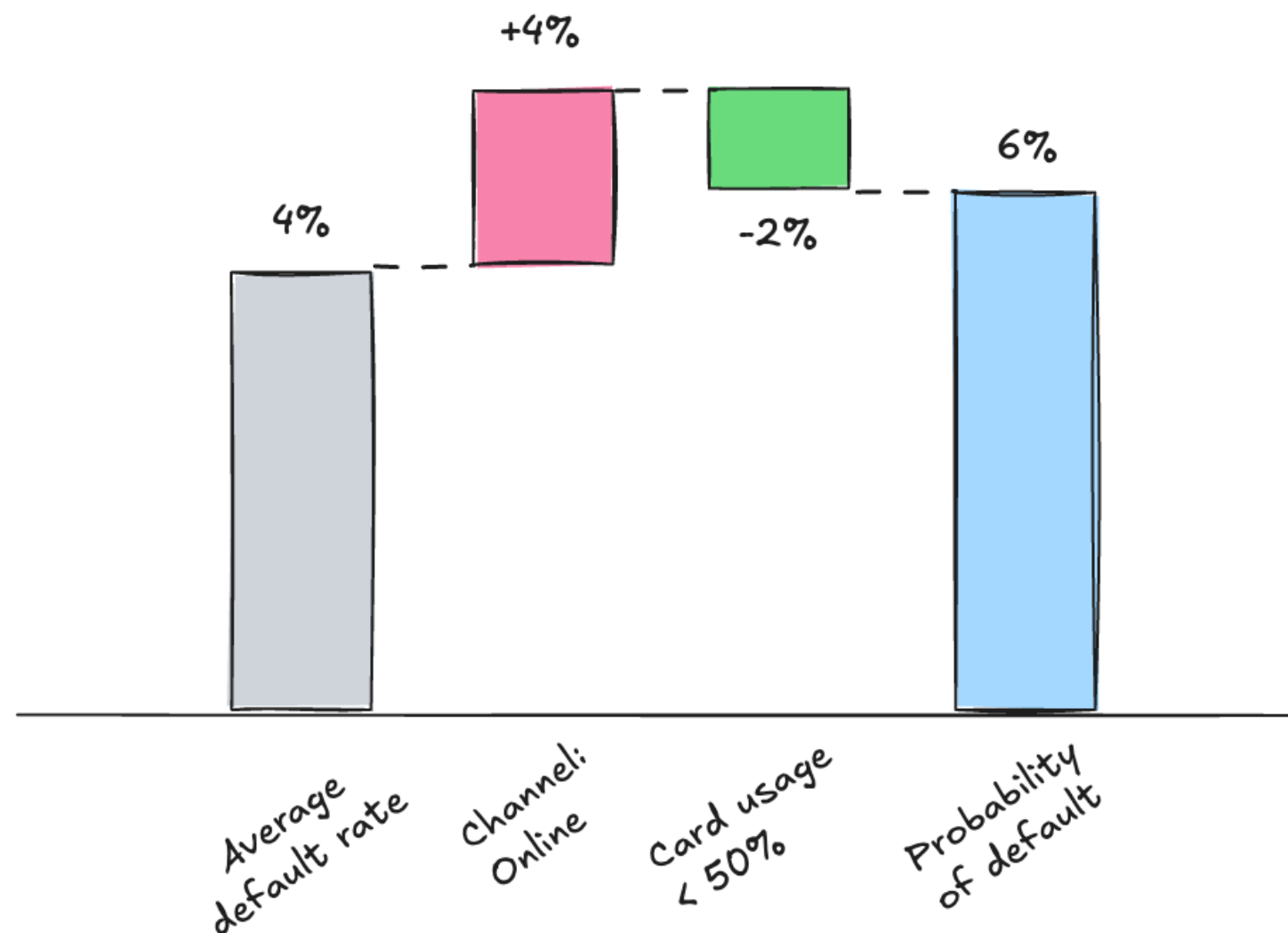


FastWoe

Fast Weight of Evidence Encoding and Inference



```
%pip install fastwoe
from fastwoe import FastWoe
X_woe = FastWoe().fit_transform(X)
```



FastWoe is a predictive modeling toolkit based on weight of evidence (WOE) transformation methods, with theoretical foundations in information theory, now standard in credit scoring

With **FastWoe** you can perform univariate screening, calculate uncertainty of WOE & IV, calculate Somers' D / Gini, perform inference with Naïve Bayes classification for binary and multiclass problems

For binning continuous features, **FastWoe** offers decision trees, histograms, and FAISS Kmeans (both CPU and GPU modes supported)

With recent additions, **FastWoe** includes new plotting batteries to draw cumulative accuracy profile (CAP) curves and perform efficient variable screening with Marginal Somers' D (MSD)

Other improvements include ultra-fast installs (total package size is under 70 KB!) and supports Python 3.9+ versions

FastWoe

Marginal Somers' D



Marginal Somers' D (MSD) selects features by their incremental rank correlation with what the current model does not explain.

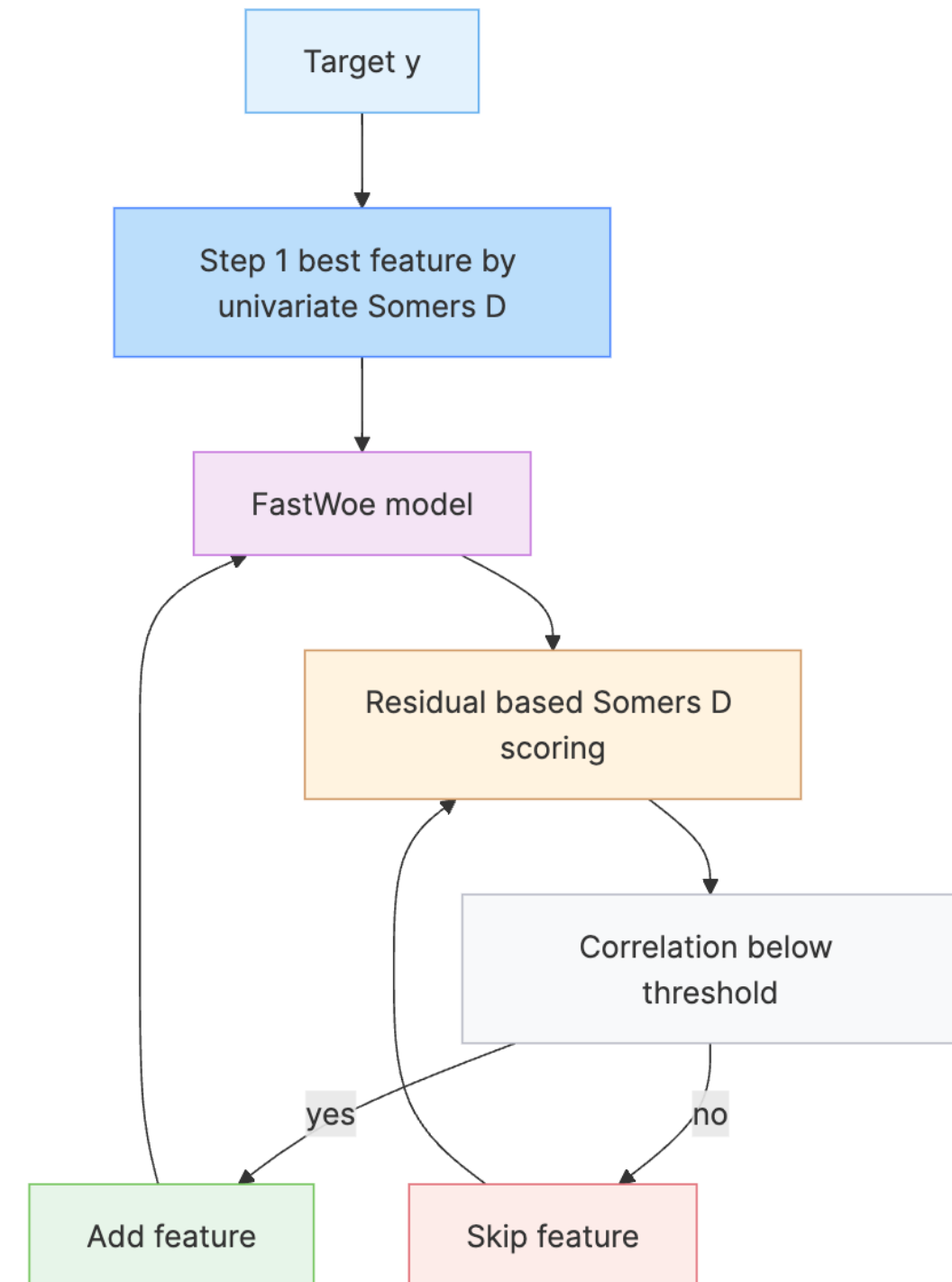
The first feature maximizes Somers' D with the target.

Each subsequent feature maximizes Somers' D with the model residuals, subject to a correlation filter.

```
from fastwoe.screening import marginal_somersd_selection

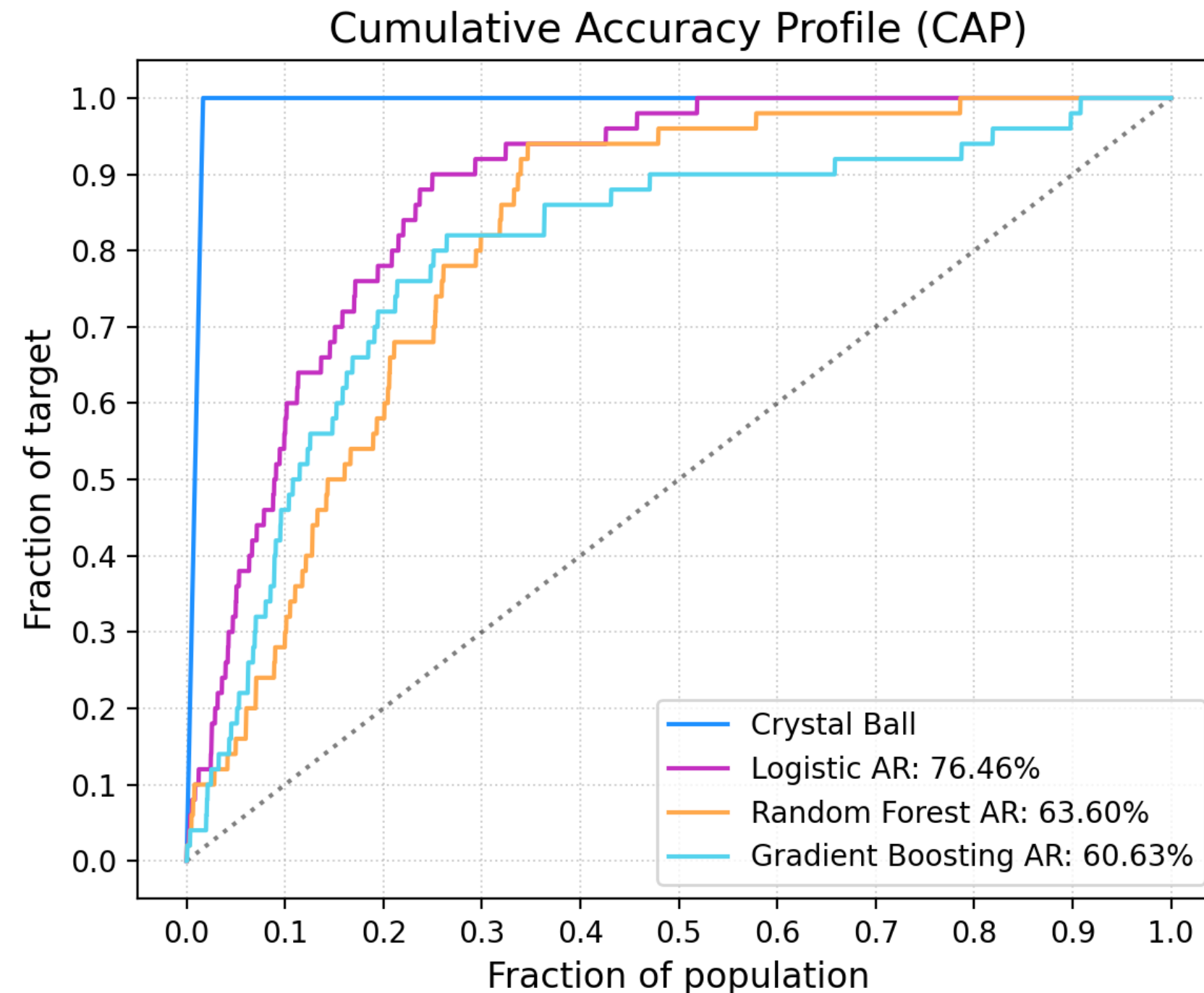
# Run MSD feature selection
result = marginal_somersd_selection(
    X_train,
    y_train,
    X_test=X_test,
    y_test=y_test,
    min_msd=0.01, # Minimum MSD threshold
    max_features=10, # Max features to select
    correlation_threshold=0.6, # Max Somers' D
    ties="y",
    random_state=42,
)

print(f"Features: {result['selected_features']}")
```



FastWoe

Cumulative Accuracy Profile (CAP) Curves



Cumulative Accuracy Profile (CAP) is a credit risk data science plotting tool to visualize discriminatory power of the model

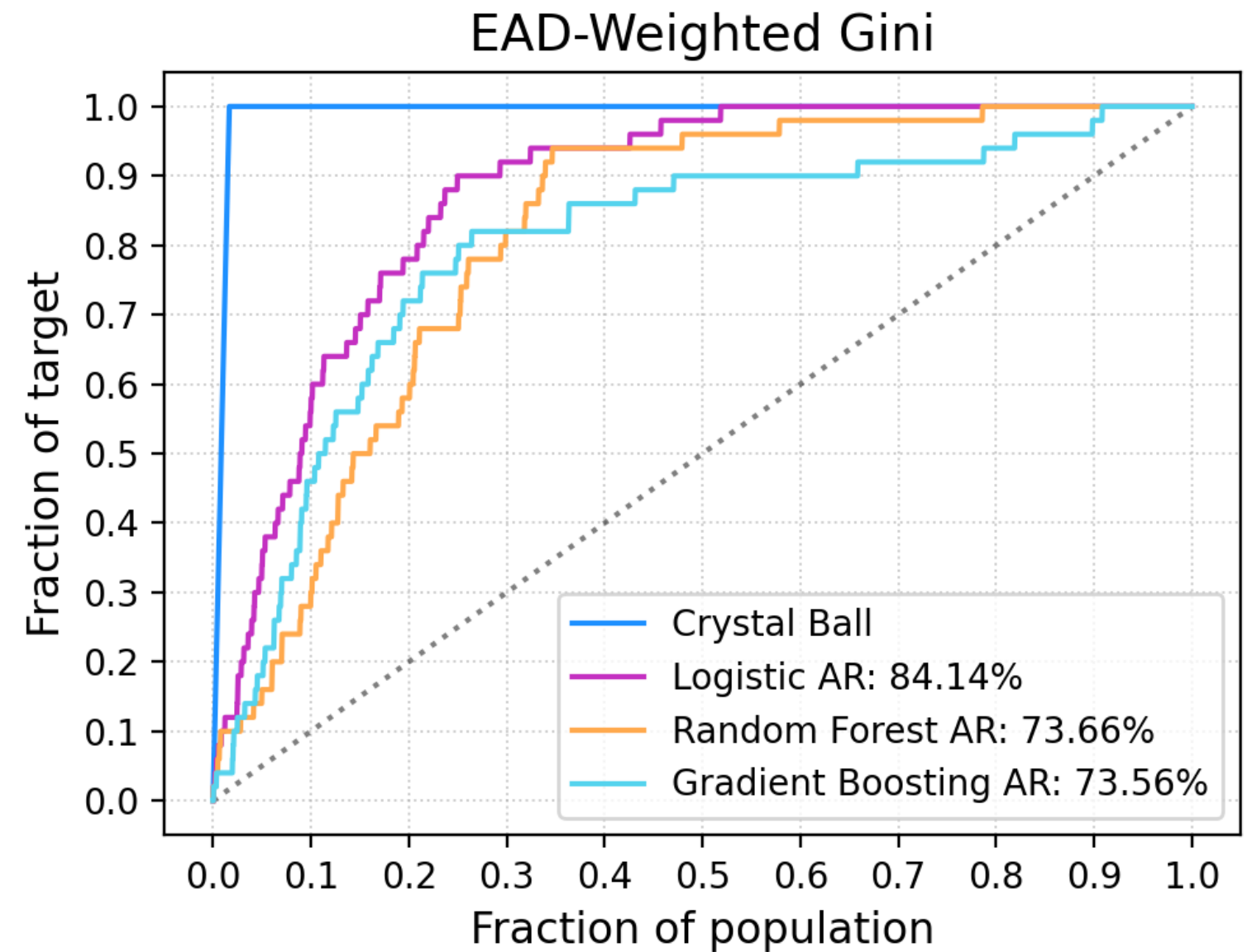
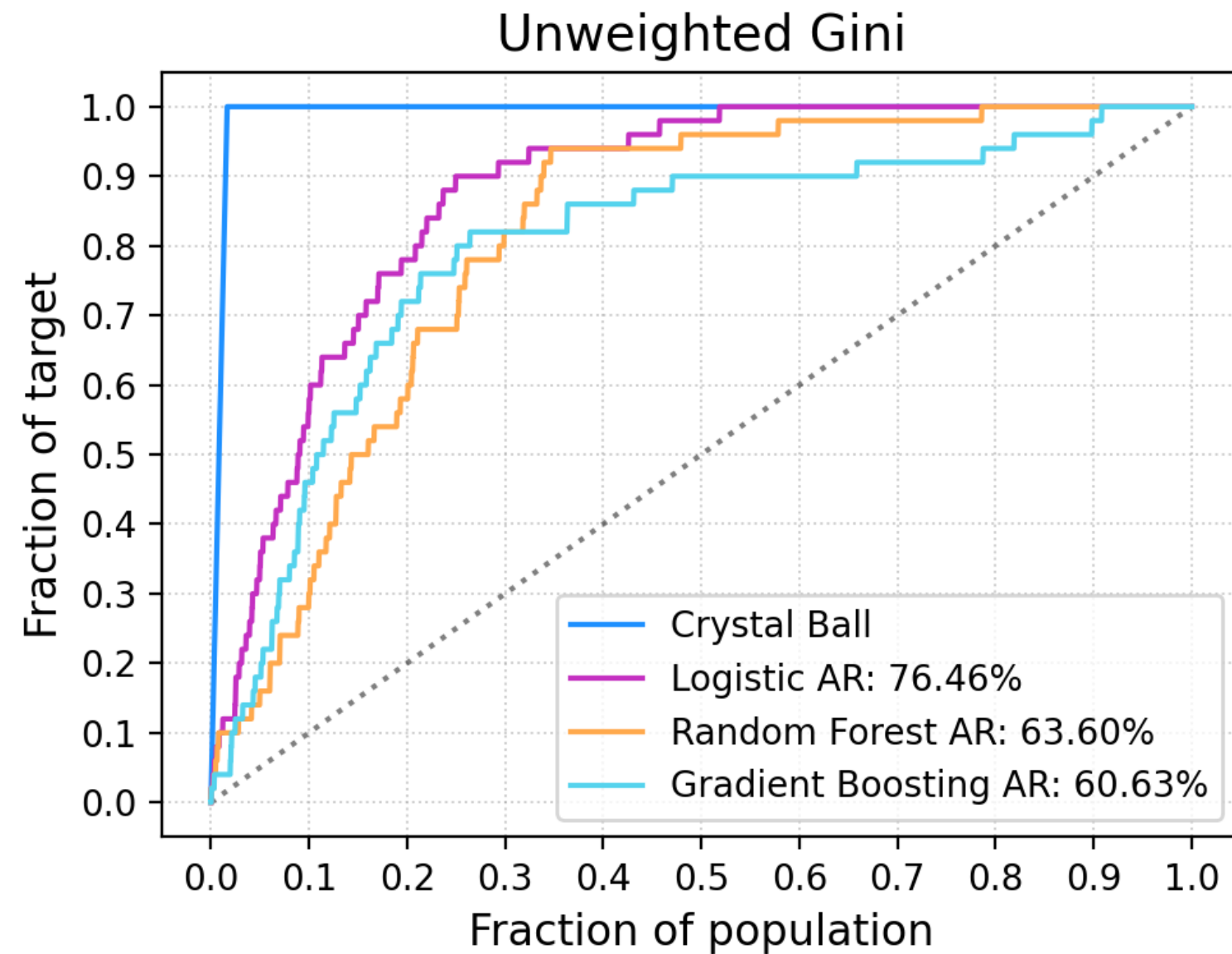
CAP allows to see the capture rate of defaults at different proportions of the scored population, the area between the model curve and ideal - random is the Accuracy Ratio (Gini, Somers' D, $AUC \times 2 - 1$)

```
from fastwoe.plots import plot_performance

fig, ax, ginis = plot_performance(
    y_true=y_test,
    y_pred=[
        predictions["Logistic"],
        predictions["Random Forest"],
        predictions["Gradient Boosting"],
    ],
    labels=[
        "Logistic",
        "Random Forest",
        "Gradient Boosting"
    ],
)
```

FastWoe

Weighted Cumulative Accuracy Profile (CAP) Curves



Cumulative Accuracy Profile (CAP) with EAD weights allow to visualize performance taking into account the monetary value of loan exposure (or any other metric that is important to business)

FastWoe

Visualization Decorators



```
from fastwoe.display import iv_styled

@iv_styled
def get_iv_analysis():
    """Get IV analysis with automatic styling."""
    return woe_encoder.get_iv_analysis()

# Automatically returns styled output
get_iv_analysis()
```

```
from fastwoe.display import styled

@styled(
    highlight_cols=["iv", "gini"],
    precision=4,
    theme="light"
)

def get_top_features_light(n=5):
    """Get top N features by IV and Gini."""
    iv_df = woe.get_iv_analysis()
    return iv_df.nlargest(n, "iv")[
        ["feature", "iv", "gini", "iv_se"]
    ]

# Get Top 5 Features
get_top_features_light(n=5)
```

Information Value Analysis

Feature importance and statistical significance for WOE encoding

FEATURE	IV	IV_SE	IV_CI_LOWER	IV_CI_UPPER	IV_SIGNIFICANCE	N_CATEGORIES	GINI
inquiries	1.8697	0.2454	1.3888	2.3507	Significant	8	0.5942
income	1.6458	0.2527	1.1505	2.1411	Significant	8	0.5293
num_accounts	0.9174	0.1618	0.6003	1.2345	Significant	8	0.4244
age	0.7046	0.1361	0.4378	0.9713	Significant	8	0.3639
loan_amount	0.5733	0.1265	0.3254	0.8212	Significant	7	0.3545
debt_to_income	0.3490	0.1453	0.0642	0.6337	Significant	6	0.2265
utilization_rate	0.2998	0.1360	0.0334	0.5663	Significant	7	0.1495
credit_history	0.1999	0.1079	0.0000	0.4113	Not Significant	5	0.1280
employment_length	0.1551	0.0827	0.0000	0.3171	Not Significant	6	0.1084
payment_history	0.1374	0.0839	0.0000	0.3018	Not Significant	7	0.0933

1. The `@iv_styled` decorator automatically applies IV analysis styling to check the variable screening.
2. The `@styled` decorator allows you to turn any DataFrame into a stylized version for highlighting the relevant points for communicating the results.